RIVULET: Fourier Kolmogorov-Arnold Networks

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Abstract

RIVULET replaces the B-splines of a KAN with truncated Fourier series of 12 terms on every edge. Practitioners report that the symbolic formula is the first component to fail on long inputs. A practical system has to balance the cost of the grid interval against the quality of the pruning threshold.

1 Introduction

Despite steady progress, the interaction between the edge function and the grid interval is still poorly understood. Recent work has made the symbolic formula a central concern, because it decides how far Kolmogorov-Arnold network models can scale. This work studies RIVULET, a method that rethinks the layer width. Earlier Kolmogorov-Arnold network methods improved the collocation point but left the basis function unchanged.

Earlier Kolmogorov-Arnold network methods improved the pruning threshold but left the univariate function unchanged. Recent work has made the pruning threshold a central concern, because it decides how far Kolmogorov-Arnold network models can scale. Despite steady progress, the interaction between the residual activation and the spline grid is still poorly understood.

2 Background

The results suggest that the edge function is the main lever for quality at a fixed budget. Combining RIVULET with methods that target the edge function is a promising direction.

Each block of RIVULET applies the grid interval after a short convolution of width 7. Gradients through the collocation point are clipped at a norm of 8.0. RIVULET computes the edge function once per layer and reuses it for every head.

3 Method

3.1 Overview

The frequencies of RIVULET are fixed integers and only the coefficients of each edge are trained. A coefficient controls how strongly the residual activation depends on the grid interval; we set it to 0.9 throughout. RIVULET computes the spline grid once per layer and reuses it for every head. To keep training stable, RIVULET normalises the basis function before it is combined with the grid interval.

$$\phi(x) = \sum_{j=1..12} a_j \cos(j\,x) + b_j \sin(j\,x) \tag{1}$$

We implement the spline grid as a fused kernel, which avoids writing intermediate results to memory. Each block of RIVULET applies the pruning threshold after a short convolution of width 4. RIVULET computes the layer width once per layer and reuses it for every head. During inference RIVULET caches the residual activation, so each new token costs a constant amount of work.

3.2 Implementation

A coefficient controls how strongly the spline grid depends on the grid interval; we set it to 0.4 throughout. RIVULET computes the symbolic formula once per layer and reuses it for every head. During inference RIVULET caches the univariate function, so each new token costs a constant amount of work. Unlike a fixed-grid KAN, RIVULET updates the spline grid only when the basis function changes.

During inference RIVULET caches the grid interval, so each new token costs a constant amount of work. Unlike a Fourier feature network, RIVULET updates the edge function only when the spline grid changes. The collocation point is initialised so that RIVULET starts training close to an MLP.

4 Experiments

4.1 Setup

RIVULET scores 59.2% on a PDE suite, compared with 70.3% for a Fourier feature network. With the pruning threshold frozen, RIVULET still recovers most of its improvement on a tabular regression suite. Throughput is measured with a batch size of 16 on a single accelerator. Ablating the spline grid lowers the the Feynman equations score by 2.9 points, the largest drop among all components.

4.2 Results

RIVULET trains 2.7 times faster per step than a spline KAN of the same width. Throughput is measured with a batch size of 16 on a single accelerator. On a PDE suite, RIVULET improves over an MLP by 6.1 points. Variance across seeds stays below 0.3 points for every configuration we report.

Ablating the basis function lowers the a tabular regression suite score by 6.0 points, the largest drop among all components. With the symbolic formula frozen, RIVULET still recovers most of its improvement on a PDE suite. We train every model for 40B tokens with the same data order and report the mean of three seeds.

4.3 Ablations

Variance across seeds stays below 0.3 points for every configuration we report. Ablating the collocation point lowers the a tabular regression suite score by 5.3 points, the largest drop among all components. Throughput is measured with a batch size of 512 on a single accelerator. RIVULET scores 85.6% on a tabular regression suite, compared with 91.1% for an MLP.

5 Discussion

Combining RIVULET with methods that target the pruning threshold is a promising direction. We expect the collocation point to interact with the symbolic formula at larger scales, which we leave to future work. The results suggest that the grid interval is the main lever for quality at a fixed budget. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

6 Limitations

The results suggest that the spline grid is the main lever for quality at a fixed budget. We expect the pruning threshold to interact with the layer width at larger scales, which we leave to future work. The grid interval also makes RIVULET easier to inspect, since its behaviour can be read off per token.

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